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WES237C - Fall26 Quarter

Question 1 - FIR11 Baseline:

- **(a)** Report latency,initiation interval (II), in clock cycles.
- **(b)** Report the number of BRAM, DSP, LUT and FFs used.

	Fir11 Baseline
Latency (in cycles)	19
Interval (in cycles)	20
BRAMs	0
DSP	2
LUTs	383
FF	733

Question 2 - Variable Bitwidths:

- **(a)**

	bitWidth16	bitWidth32	bitWidth64
Latency (in cycles)	134	135	135
Interval (in cycles)	135	136	136
BRAMs	2	2	2

DSP	1	2	2
LUTs	239	315	330
FF	244	588	622

- (b)
 - Minimum bitwidth is 16, bitwidth of 8 failed to match golden output

Question 3 - Pipelining:

- (a)

	Fir128 Baseline
Latency (in cycles)	135
Interval (in cycles)	136
BRAMs	3
DSP	2
LUTs	305
FF	610

- (b)

	Fir128 Baseline Pipeline Off
Latency (in cycles)	Min: 385 max:513
Interval (in cycles)	Min: 386 max: 514
BRAMs	1

DSP	1
LUTs	198
FF	127
Throughput	0.2989 MHz

- **(c)**

	Fir128 Baseline Pipeline II=1	Fir128 Baseline Pipeline II=2	Fir128 Baseline Pipeline II=3	Fir128 Baseline Pipeline II=4	Fir128 Baseline Pipeline II=5
Latency (in cycles)	134	261	388	515	515
Interval (in cycles)	135	262	389	516	516
BRAMs	2	1	1	1	1
DSP	1	1	1	1	1
LUTs	239	237	254	228	228
FF	244	163	163	152	152
Throughput	1.1036 MHz	0.5687 MHz	0.3820 MHz	0.2887 MHz	0.2887 MHz

- **(d)**

- **At a pipeline value of 4 we seem to reach a limit value. The resource usages and throughput between the two instances do not change**

- **(e)**

- **The value of the automatic pipeline may be near 4 or 5?**

Question 4 - Removing Conditional Statements:

- (a)

	Fir128 No Conditionals Auto Pipeline	Fir128 With Conditionals Auto Pipeline
Latency (in cycles)	134	135
Interval (in cycles)	135	136
BRAMs	1	3
DSP	1	2
LUTs	214	305
FF	130	610
Throughput	1.1036 MHz	1.0638 MHz

- (b)

	Fir128 No Conditionals Pipeline Off	Fir128 With Conditionals Pipeline Off
Latency (in cycles)	134	Min: 385 max:513
Interval (in cycles)	135	Min: 386 max: 514
BRAMs	1	1
DSP	1	1
LUTs	214	198
FF	130	127
Throughput	1.1036 MHz	0.2989 MHz

Question 5 - Loop Partitioning:

- (a)
 - Here I partition the loops by breaking them into a shift reg handling loop and another for the accumulation loop. This allows us to run in parallel
- (b)

	With Loop Partitioning	Without Loop Partitioning (Baseline)
Latency (in cycles)	267	135
Interval (in cycles)	268	136
BRAMs	1	3
DSP	1	2
LUTs	273	305
FF	128	610
Throughput	0.5559 MHz	1.0638 MHz

- (c)

	Partitioned and Unrolled Manually	Partitioned and Unrolled w/ Pragmas
Latency (in cycles)	133	253
Interval (in cycles)	134	254
BRAMs	6	6
DSP	4	36

LUTs	1529	8550
FF	1224	10937
Throughput	1.1467 MHz	0.5642 MHz

I ran the experiment with a manually unrolled version of the loop, as well as with the pragmas, the manual performs better in terms of increased throughput, fewer resources, and lower latency.

- (d)
 - By running the Loop Unrolling with an auto pipeline , It looks like combining the two gives us our best throughput so far , if the experiment from Q3 was correct.

Question 6 - Memory Partitioning:

Note: I am using manual loop unrolling moving forward! (This was a mistake?)

- (a)

	Unrolled_Loop_Partitioned Complete	Unrolled_Loop_Partitioned Cyclic	Unrolled_Loop_Partitioned Block
Latency (in cycles)	106	133	133
Interval (in cycles)	107	134	134
BRAMs	0	6	6
DSP	4	4	4
LUTs	6340	1529	1529
FF	8509	1224	1224
Throughput	1.3758 MHz	1.1467 MHz	1.1467 MHz

- (b)

	Rolled_Loop_Partitioned Complete	Rolled_Loop_Partitioned Cyclic	Rolled_Loop_Partitioned Block
Latency (in cycles)	265	267	267
Interval (in cycles)	266	268	268
BRAMs	0	1	1
DSP	1	1	1
LUTs	6161	273	273
FF	3302	128	128
Throughput	0.5473 MHz	0.5559 MHz	0.5559 MHz

The loop unrolled vs rolled has a big impact on the performance of the fir filter. The memory partitioning alone is not enough to increase performance or throughput beyond the limitations of the loop.

Question 7 - Best Design:

- (a)

By combining Loop Unrolling (with Pragmas!), Partitioned, with Memory Partitioning and Pipeline II=1, I can achieve the best design. The biggest difference is that pipelining was applied to the top-level, not just the loops

	Best Design
Latency (in cycles)	27
Interval (in cycles)	1

BRAMs	0
DSP	1
LUTs	4248
FF	8703
Throughput	151.4463

- (b)

The resource usage is much higher to account for more calculations per cycle. The read write instruction count increases with the factor of loop unroll. We have also split our loop into two, which allows for simultaneous execution. This will also require more resources.

Q8 Github Link: <https://github.com/ldurb/Wes237C>